

ArcGIS Experience Builder: An Introduction

Jianxia Song, Kenyon Huber-Wilker



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2021 ESRI USER CONFERENCE

agenda



Gallery

Explore apps built with ArcGIS Experience Builder

Grid

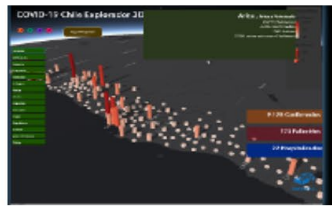
Filter

Items: 27



PBES Map Explorer

A one-stop platform for exploring maps and data from the Napa County Planning, Building, and Environmental Services Dept.



COVID-19 Chile Explorer

Explore COVID-19 statistics in Chile with this 3D explorer. Includes confirmed and active cases.



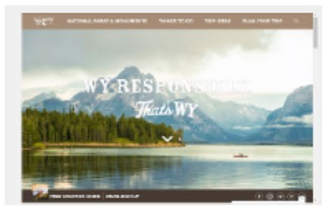
Wildfire Initial Attack Reporting

Designed for admins and managers of the wildfire initial attack feed, this is a one-stop-shop for the public map, admin application, and survey.



Switzerland COVID-19 Tracker

This application visualizes COVID-19 submissions from covidtracker.ch, including all-time and those in the last 24 hours.



Wyoming Tourism

Explore national parks resources and tourism information across the state of Wyoming.



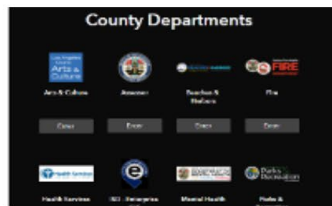
CDC Places Project

This application provides interactive maps for model-based chronic disease related estimates of the CDC PLACES (Population Level Analysis and Community Estimates) project.



Washoe County Polling Location Wait Times

This application shows the estimated wait times at Polling Locations in Washoe County, NV.



LA County GIS Day Virtual Exhibit Hall

Explore different departments and sponsors in this virtual exhibit hall, created by LA County for GIS Day 2020.



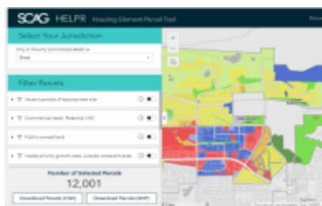
What is a Digital Twin

This app from Esri Spain discusses the idea of the Digital Twin, its use cases, and how it can be created and utilized in ArcGIS.



AmeriGEOSS Symposium & UN GGIM Americas Event

The AmeriGEOSS Symposium, a live 3-day event, was hosted through this Experience Builder website.



SCAG Housing Element Parcel Tool

HELPR is a web-mapping tool developed by SCAG to help local jurisdictions and stakeholders understand local land use, site opportunities, and environmental sensitivities for housing planning.



Maryland DOT SHA Context Driven

The MDOT SHA Context Driven guide is a planning and design resource offering practitioners guidelines centered on establishing safe and effective multi-modal transportation systems.

What is ArcGIS Experience Builder

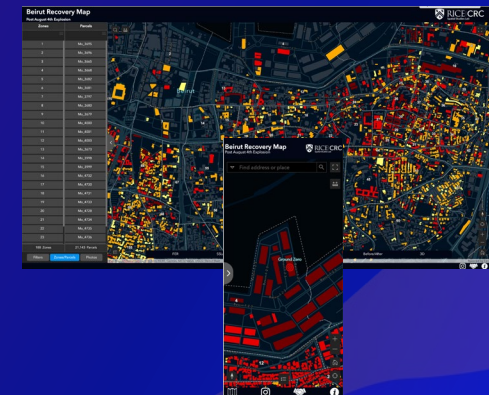
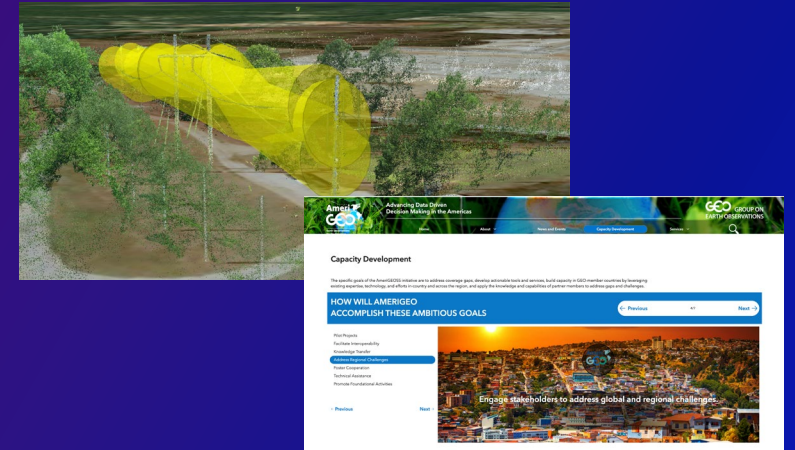
ArcGIS Experience Builder enables you to create web apps and pages without coding

- **Use a template or start from scratch to create exact web experience you want.**

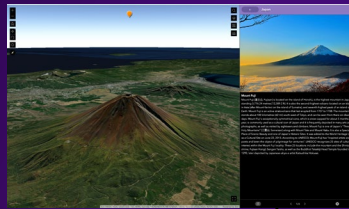
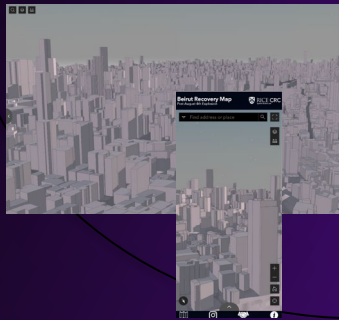
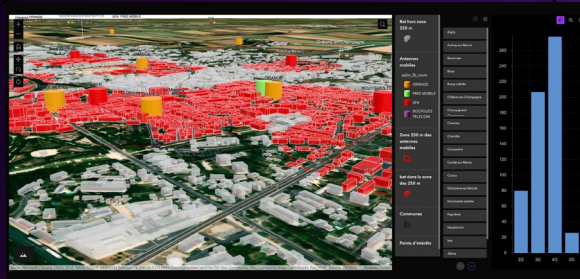


ArcGIS Experience Builder Key Features

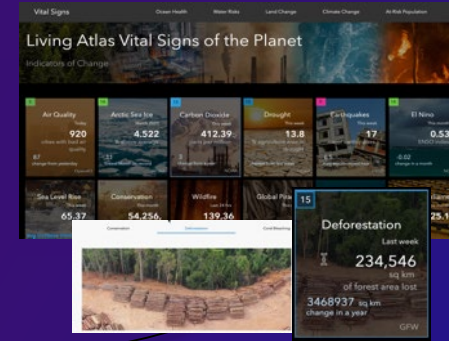
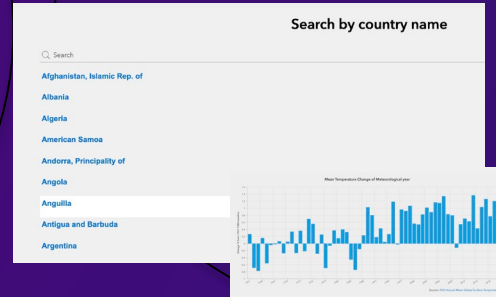
- **Flexible layouts**
 - Design your own blueprint
- **Mobile optimization**
 - Choose widgets to be displayed only on the desktop but not on the mobile
- **Integration**
 - 2D and 3D content into one app
 - Integrate with other apps to streamline the process
- **Interconnection**
 - Widgets communicate with each other and respond accordingly
- **A rich set of templates and widgets**



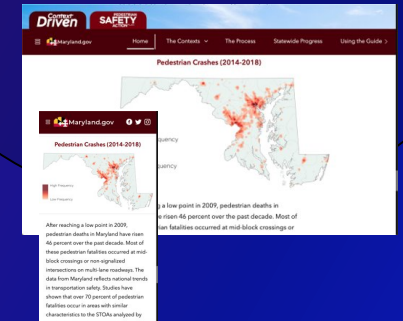
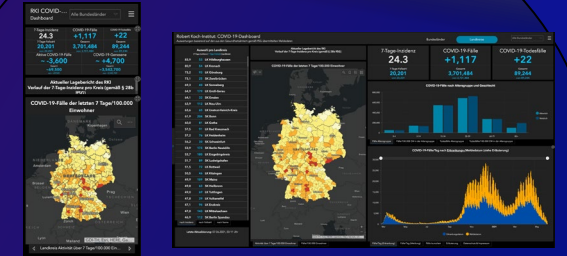
3D



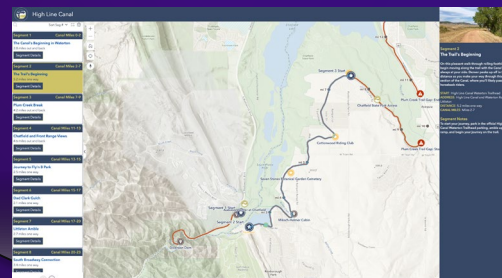
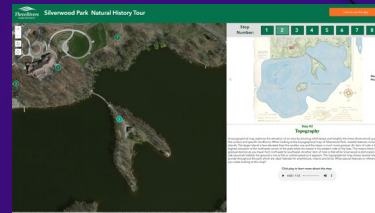
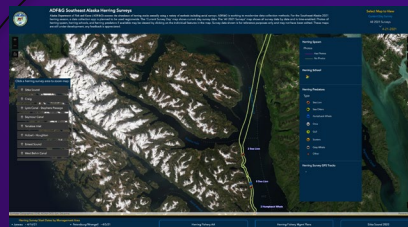
Web pages



Mobile



Classic apps



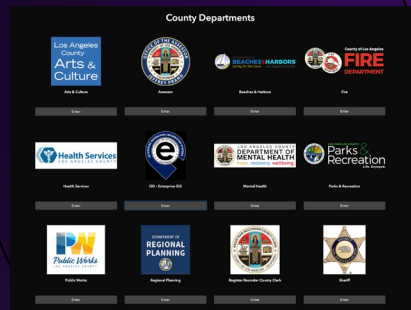
Survey

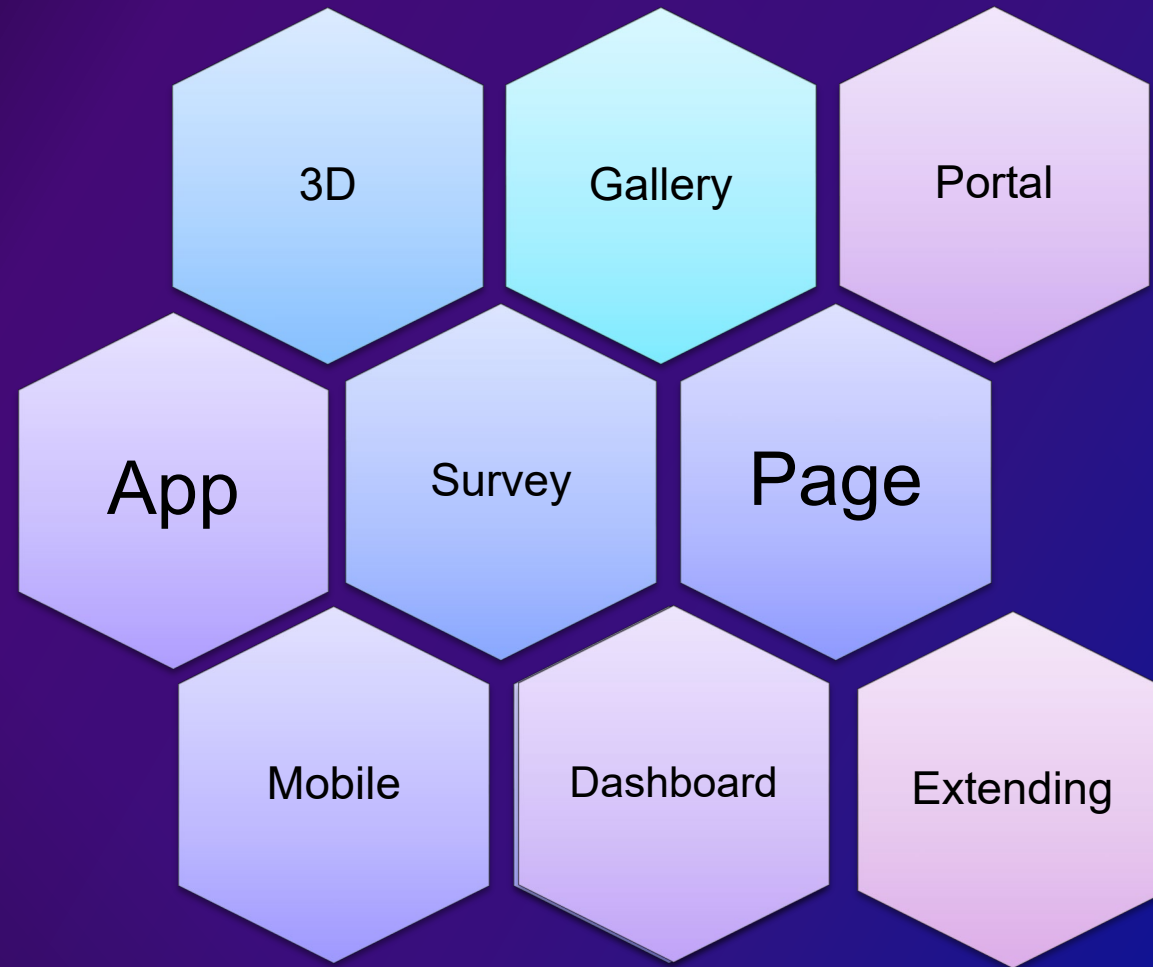
A screenshot of a survey form titled "Please fill the following form". It includes a search bar, a list of results, and a detailed view of the form.

Portal



Gallery





Limitless Creativity

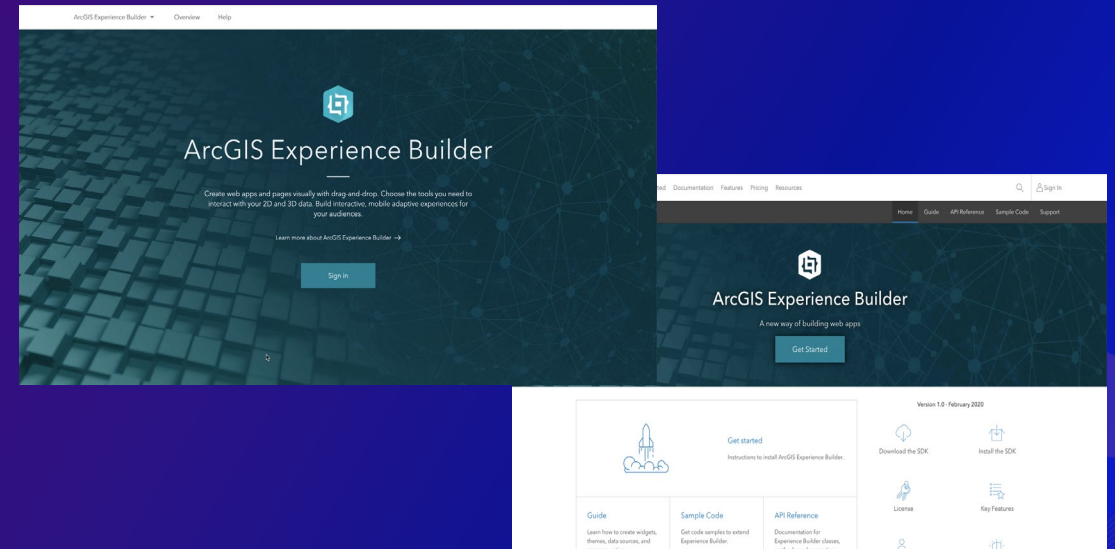
ArcGIS Experience Builder Editions

- ArcGIS Online and ArcGIS Enterprise

- URL: experience.arcgis.com
- ArcGIS Enterprise 10.8.1 and up

- Developer Edition

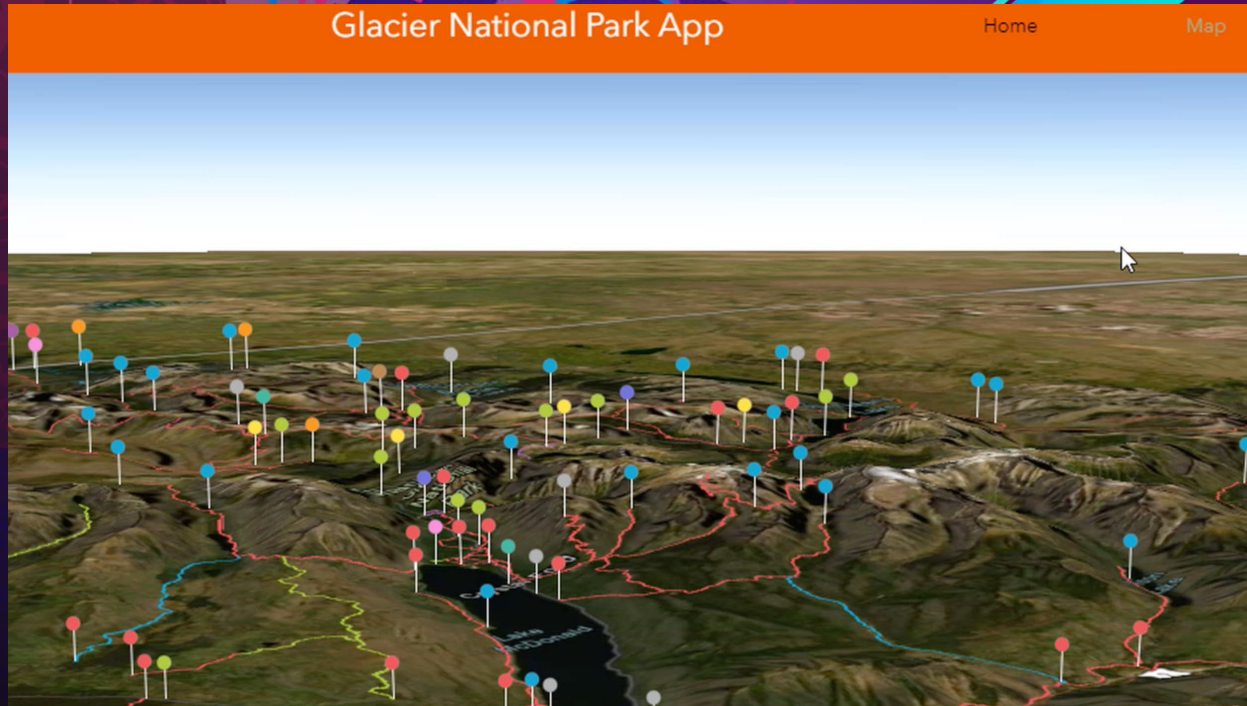
- Download from the Developers website and install locally
- Sign in to ArcGIS Online / Enterprise



ArcGIS Experience Builder Licensing



- **As part of the Essential Apps Bundle**
 - **Must have a Creator or GIS Professional user type to create experiences**
- **Through an ArcGIS Developer subscription**
 - **Allows developers who are not part of an ArcGIS organization to access and use Experience Builder Developer Edition**



Quick Tour

Kenyon Huber-Wilker

An
experience

Design/
config

GUI builder



Templates

01



Widgets

02



Data sources

04



Themes

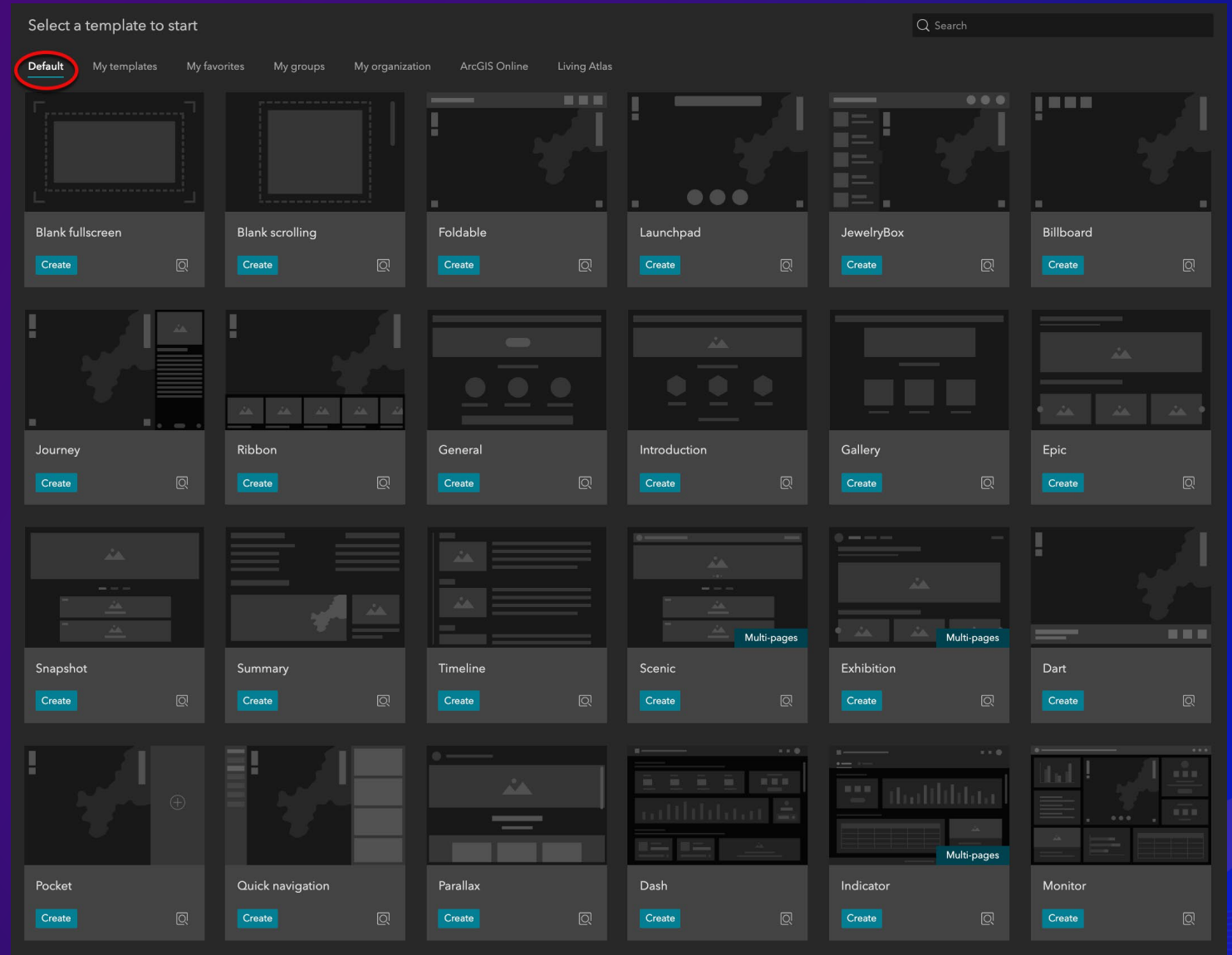
03

Template

Building Block

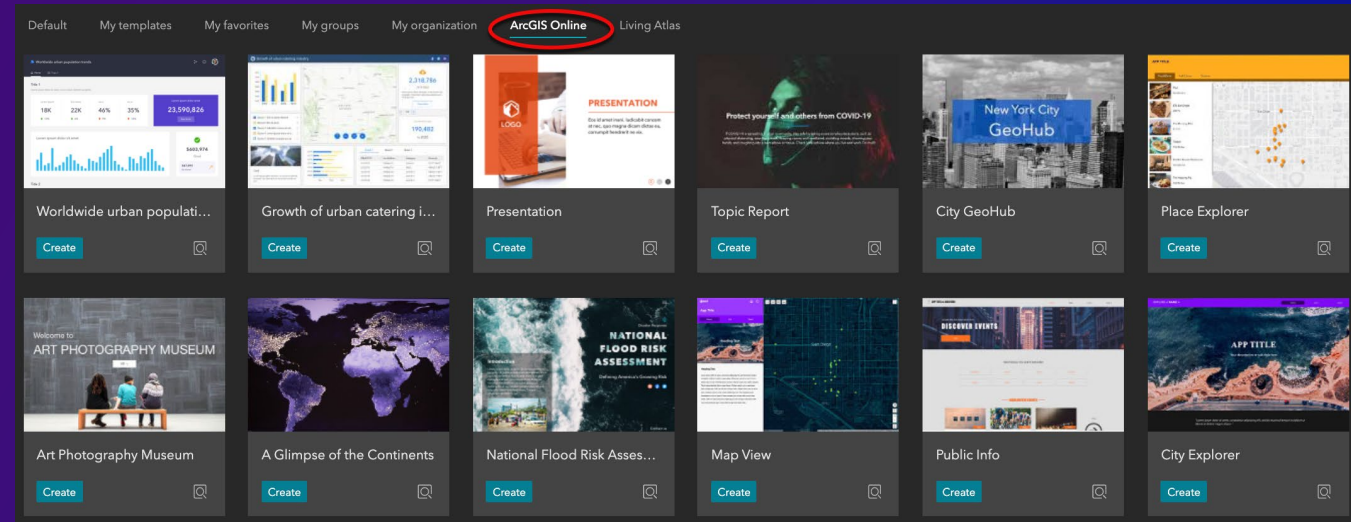
Default Templates

- **Full Screen**
 - Blank
 - Foldable
 - Launchpad
- **Scrolling Page**
 - Blank
 - Monitor **New**
 - Quick Navigation **New**
 - Indicator (multi-page) **New**
 - Exhibition (multi-page)
- **Part of software**



Other Templates

- Created by you or others
 - ArcGIS Online
 - Living Atlas
 - My Organization
 - My Group
 - My Favorite
 - My Template



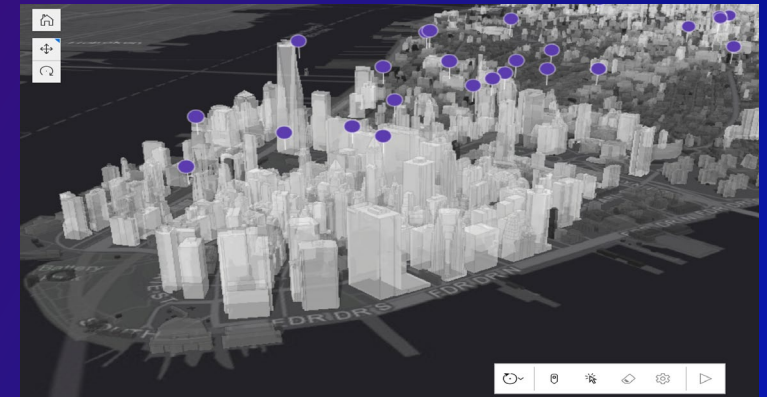
Widget

Building Block

Basic Widgets

- Map
- List
- Table
- Map Layers
- Legend
- Filter
- Query New
- Chart New
- Bookmark
- Feature Info
- Survey
- Card
- Fly Controller
- Coordinate Conversion New
- Branch Version Management New
- Share
- Widget Controller
- Button
- Text
- Image
- Embed
- Menu
- Divider

Content	Style	Action
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Which place do you enjoy the most?

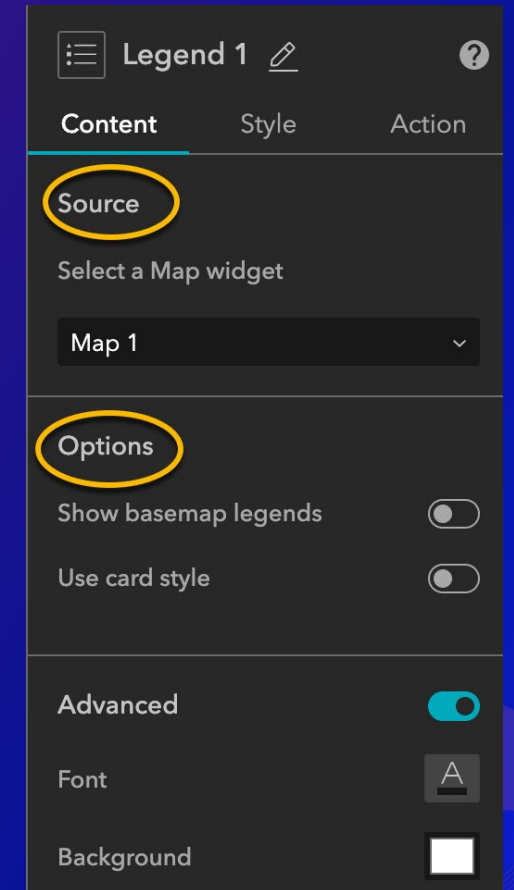
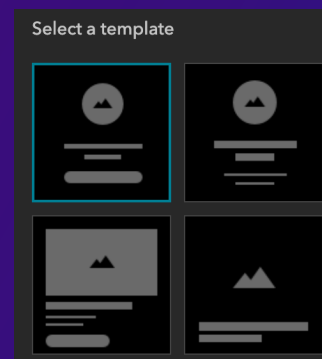
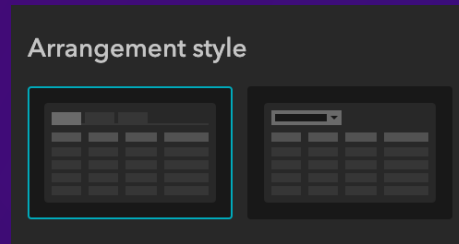
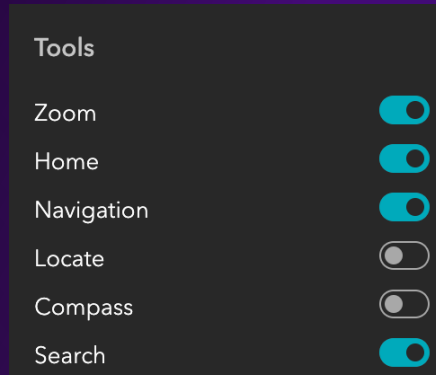
☐ Great Wall of China	☐ Mount Fuji	☐ Bali
☐ Eiffel Tower	☐ Berlin Wall Memorial	☐ Leaning Tower of Pisa
☐ Other		

Submit

Company Name	
	Bud Light Tier:1 Price:\$10 Size:16 oz
	Stella Artois Tier:2 Price:\$11 Size:16 oz
	Shocktop Tier:1 Price:\$10 Size:16 oz
	Goose Island Tier:1 Price:\$10 Size:16 oz

Widget Property - Content

- Data source
- Content options
 - Including tools, arrangement, templates, advanced options

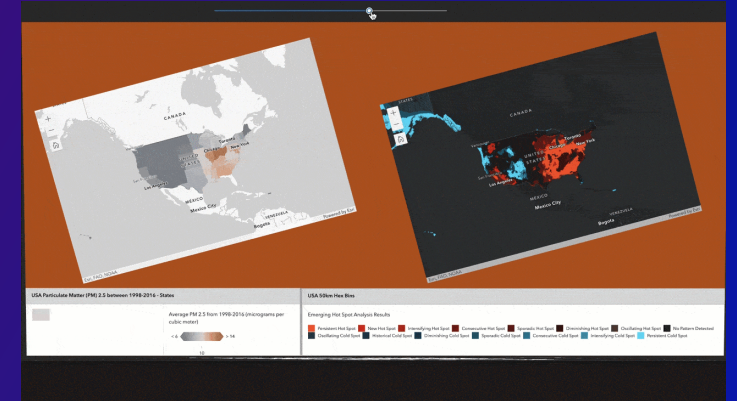
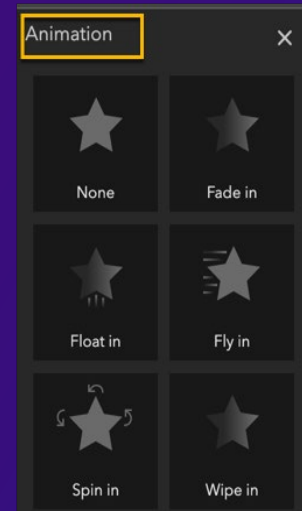
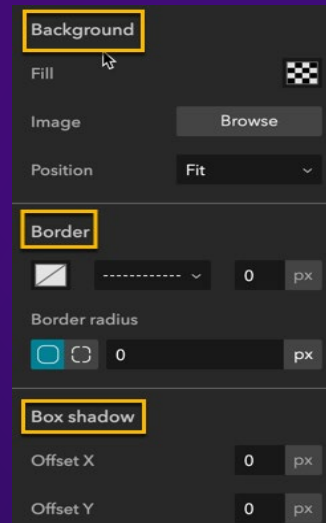
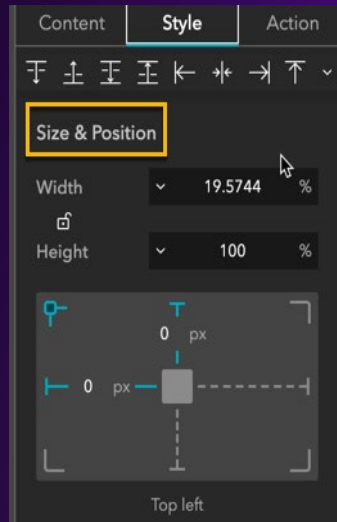


Widget Property - Style

Size & Position

Background, Border & Box shadow

Animation



Widget Property- Action

- Action makes widgets talk with each other
 - Data action **New** provides options for end users to click on like Export to CSV
 - Message action applies automatically at runtime

Interior Landmarks

BoroughID	
BX	
MN	
MN	LP-01635
MN	LP-01746
MN	LP-01096

Export to JSON
Export to CSV
Export to GeoJSON
Export all
Export selected
Zoom to
Pan to
Show on map

Data action

Table 1

Content Style Action

Message action Data action

Enable data action ⓘ ☒

Please choose the following data actions

☒ ☒ ☒ Pan to

☒ ☒ Show on map

☒ ☒ Zoom to

☐ Export ⓘ

Message action

Map 2

Content Style Action

Message action Data action

Add a trigger

Extent Changes ⓘ

Map 1

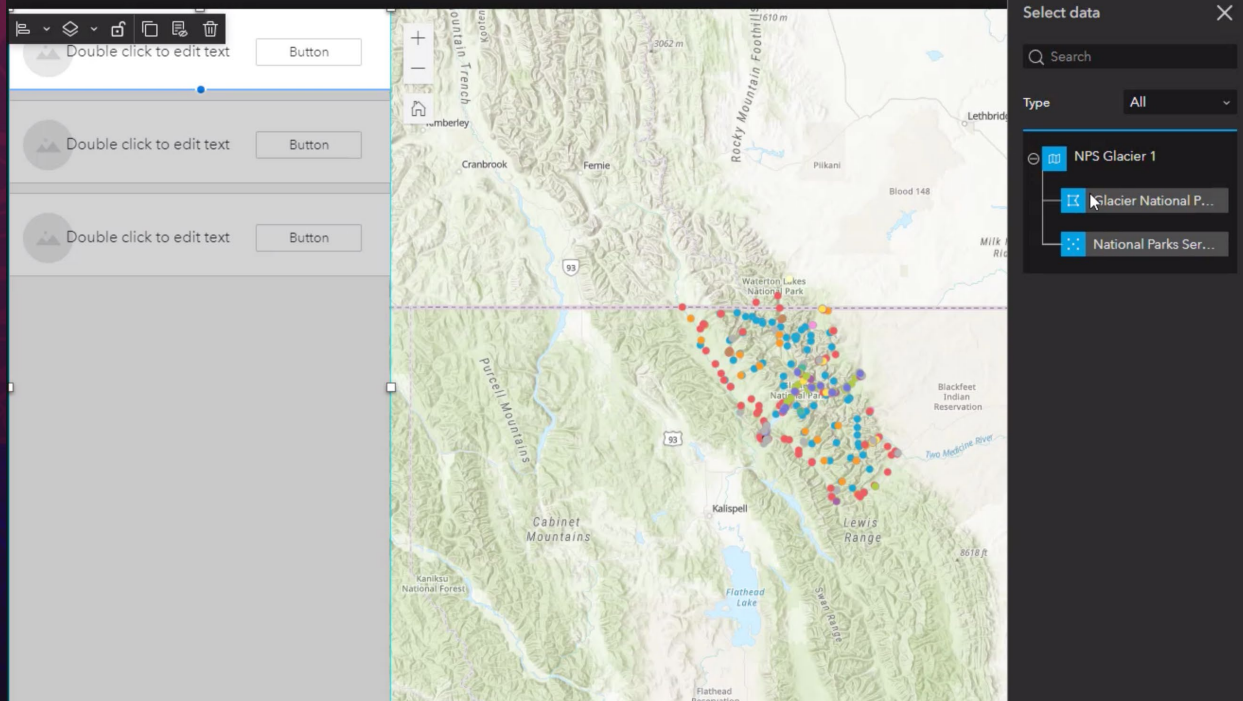
Zoom to

+ Add action

Message Action

- Consists of three parts
 - Triggers from the source widget
 - Targets that respond to the trigger, including widgets and the app framework (data)
 - Actions from the target related to its intended use from widgets

Trigger	Target	Action
Extent Changes from Map 1	Map 2	Zoom to the same extent as Map 1
	Framework (data)	Filter Data Records For example, List 1 displays only features within the current map extent



Widget Demo

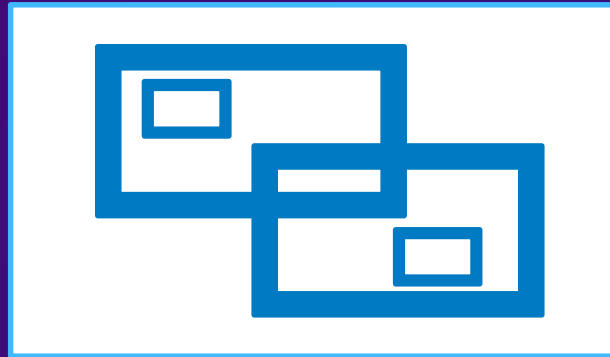
Kenyon Huber-Wilker

Layout Widgets

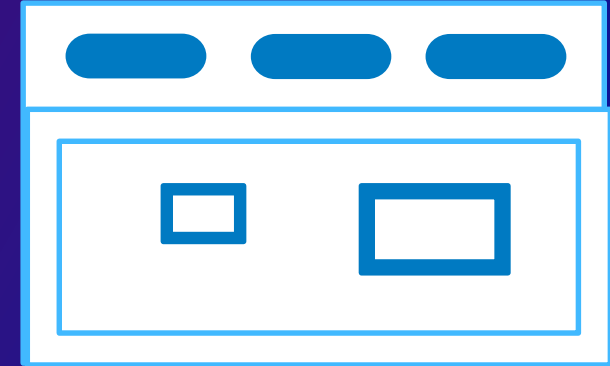
Row



Fixed Panel



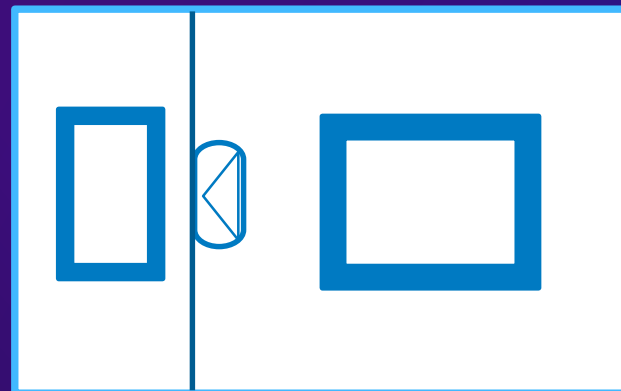
Section & View Navigation



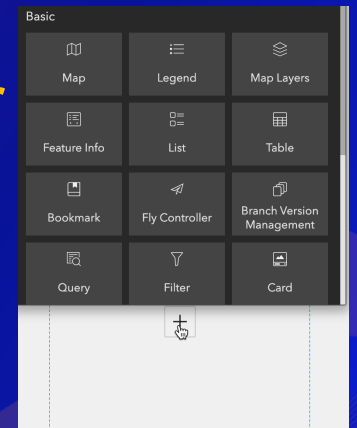
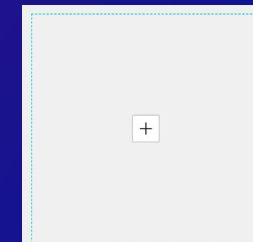
Column



Sidebar



Placeholder

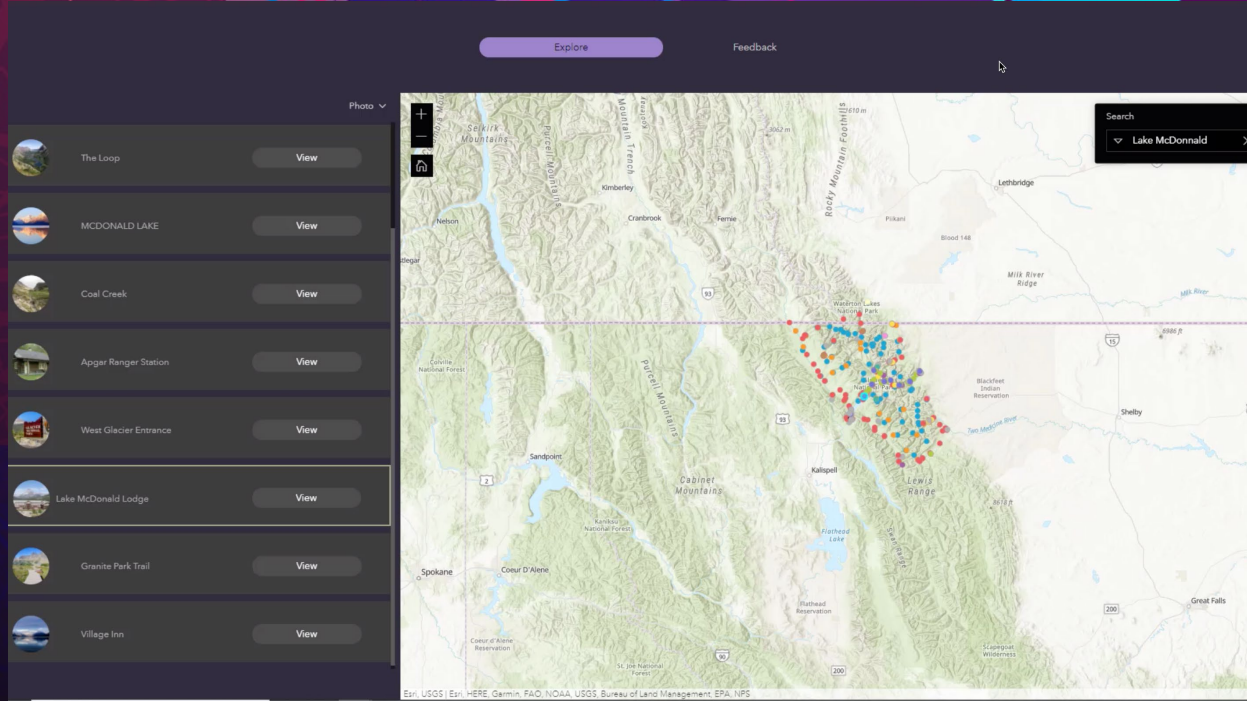


Technical session

“ArcGIS Experience Builder: Designing apps with Style and Layout”

Session ID:13164





Section Demo

Kenyon Huber-Wilker

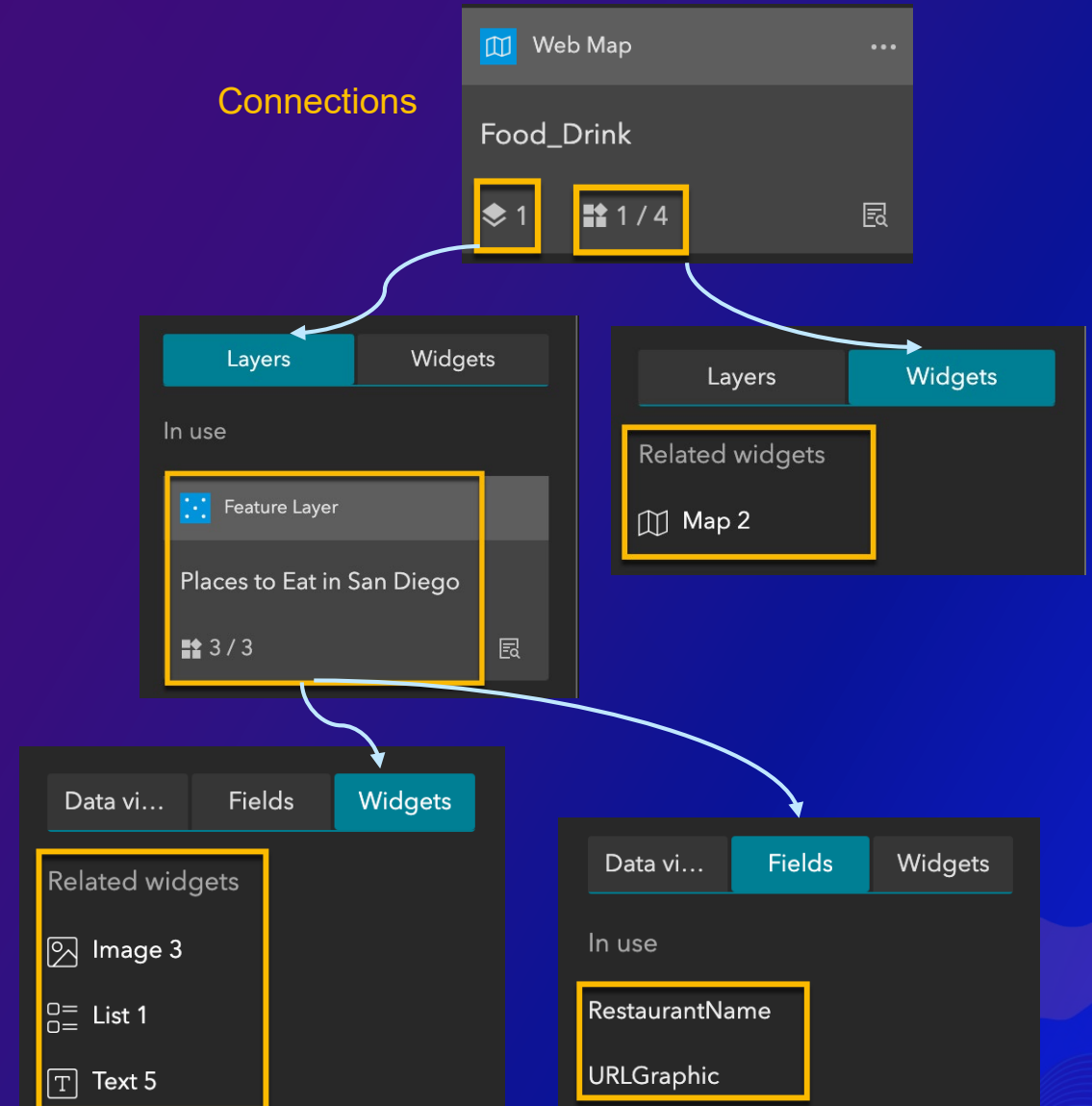
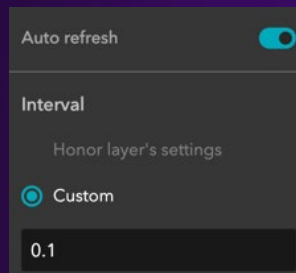


Data Source

Building Block

Data source

- Shared at the app (framework) Level
 - Web map/scene
 - Feature layer
 - Scene layer
 - Feature collection
- Show connections between layers, widgets, and fields
- Auto refresh with intervals



Data source views

- **Default view**

- A built-in view contains all the features

- **Selected features' view**

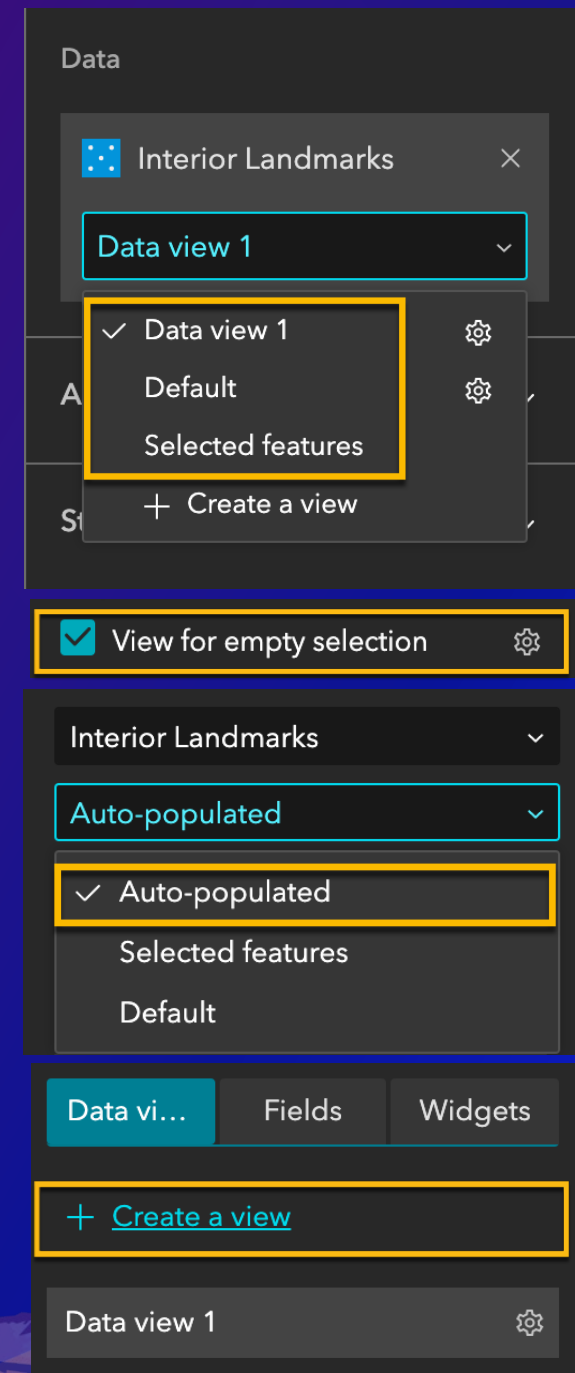
- A built-in view contains features selected by the user at runtime
- Use “**View for empty selection**” option to choose a feature to display when there's no feature selected yet

- **Auto-populated view**

- A built-in view contains one feature and is only available in the List widget

- **Data view**

- A data view is a subset of feature layer created through filtering, sorting, and records limitation
- You can have multiple data views to control which features appear in widgets



Data view 1



Filter

Sort

Records



+ Add a sort field

Distribute ...	GlobalID	website	In Places	Notes	OBJECTID	pho
Public	5b146479...		Yes	Park provi...	1	
Public	7177d821...		Yes	Park provi...	2	
Public	769c412b...		Yes	Park provi...	3	
Public	66ccd7cc...		Yes	Park provi...	4	
Public	660e80a2...		Yes	Park provi...	5	
Public	50ff3544...		Yes	Park provi...	6	
Public	41fe19aa...		Yes	Park provi...	7	
Public	663ae214...		Yes	Park provi...	8	
Public	4b2d335c...		Yes	Park provi...	9	
Public	0338e6a1...		Yes	Boat Ramp	10	
Public	199c3e7d...		Yes	Boat Ramp	11	
Public	b0441111...		Yes	Restroom	12	
Public	2b8e87f5...		Yes	Restroom	13	
Public	0078b0ff...		Yes	Picnic Area	14	

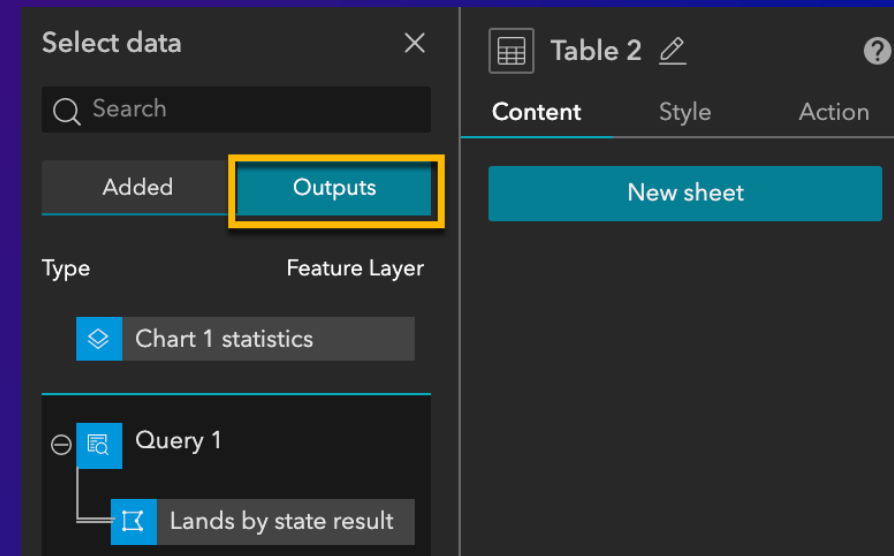
Data View Demo

Kenyon Huber-Wilker

Data source

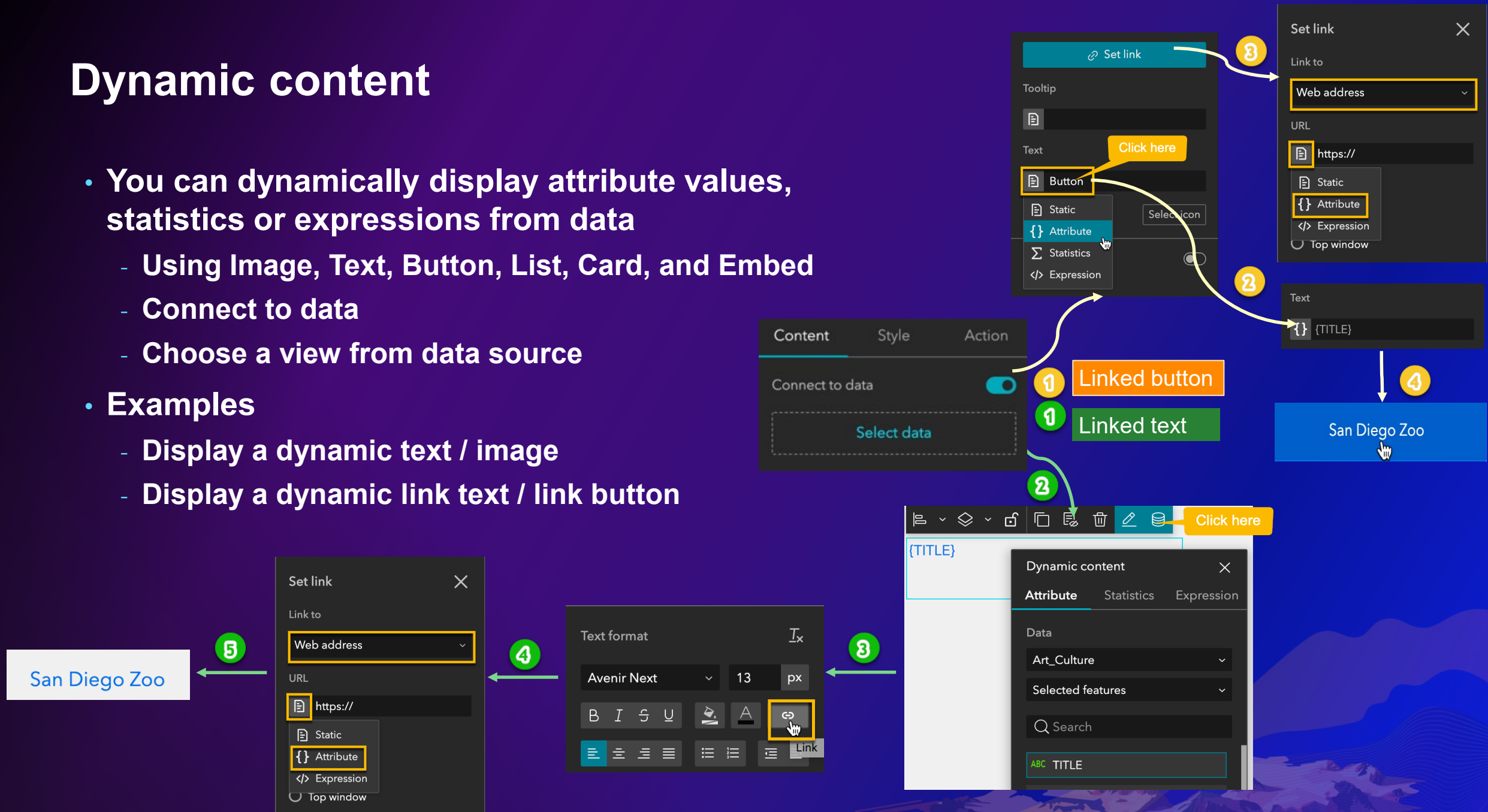
- The output data generated by widgets
 - Can be the data source to other widgets like Table, Text
 - Generated at runtime

Source widget	Output data type	Output data number
Chart	Feature layer	1
Query	Feature layer	1 for each query



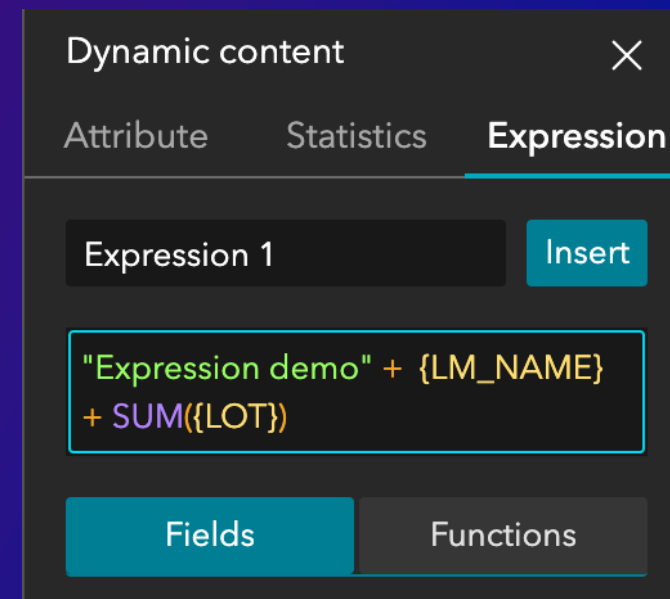
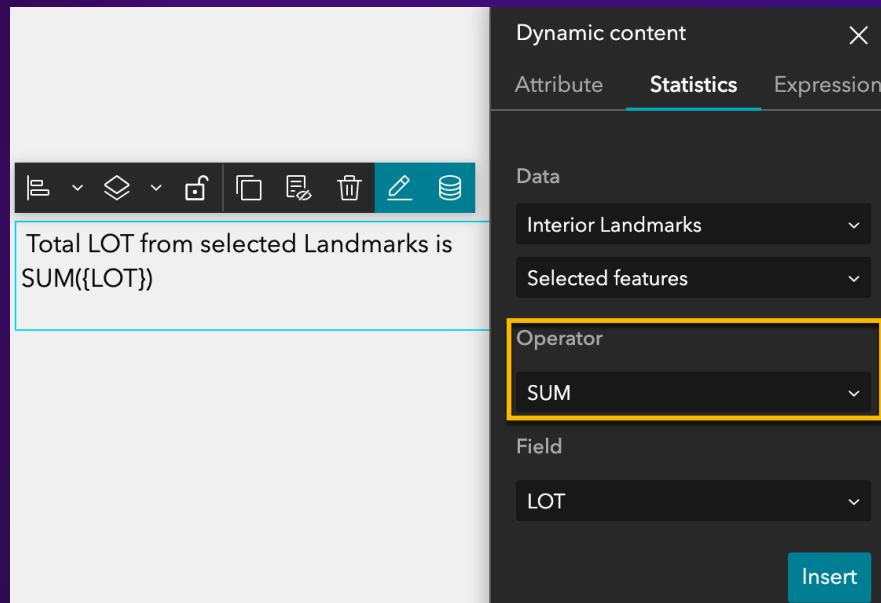
Dynamic content

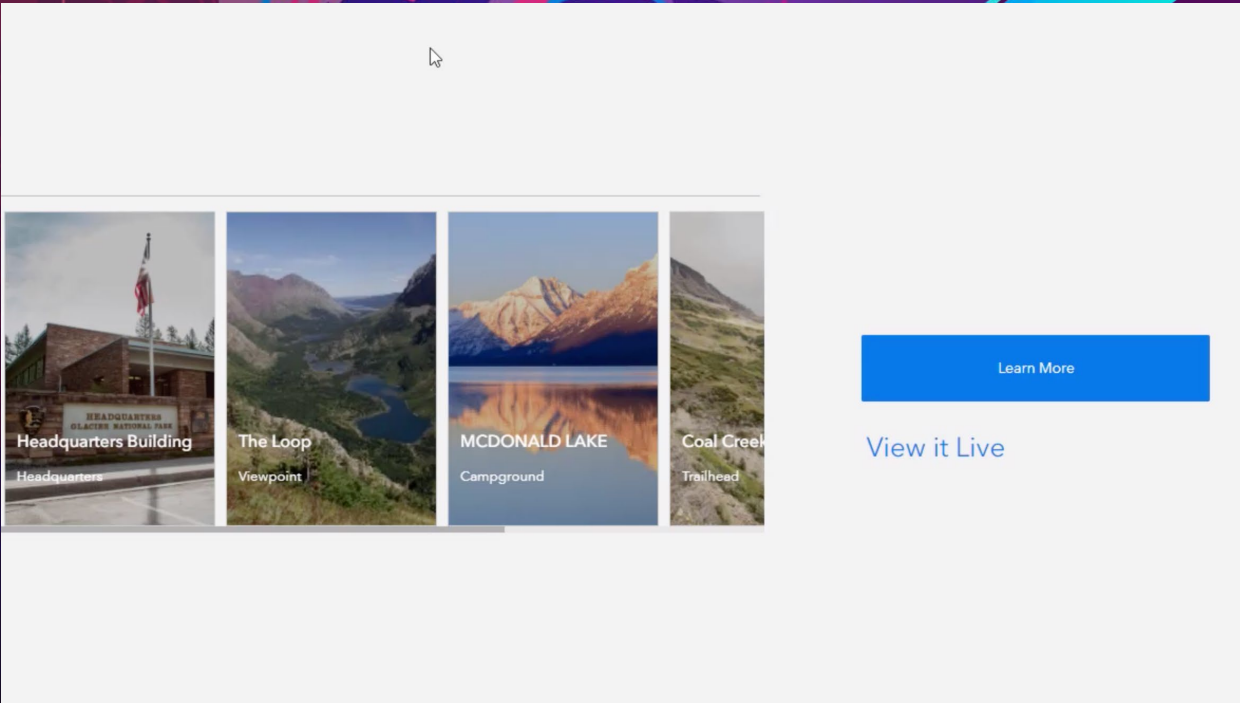
- You can dynamically display attribute values, statistics or expressions from data
 - Using Image, Text, Button, List, Card, and Embed
 - Connect to data
 - Choose a view from data source
- Examples
 - Display a dynamic text / image
 - Display a dynamic link text / link button



Dynamic content

- Examples
 - Get the statistic result of a numeric field
 - maximum, minimum, average, or sum value
 - Use the expression editor to input text, numbers, fields, and functions





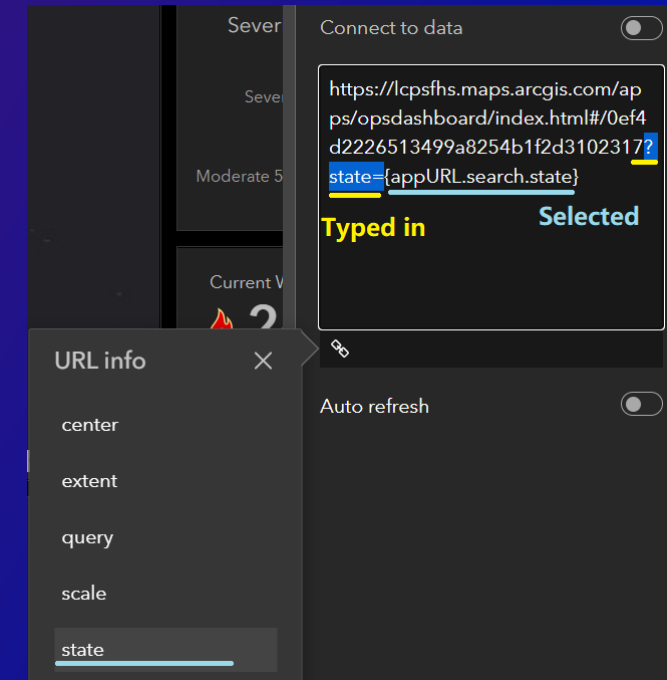
Dynamic Content For Linked Button and Text

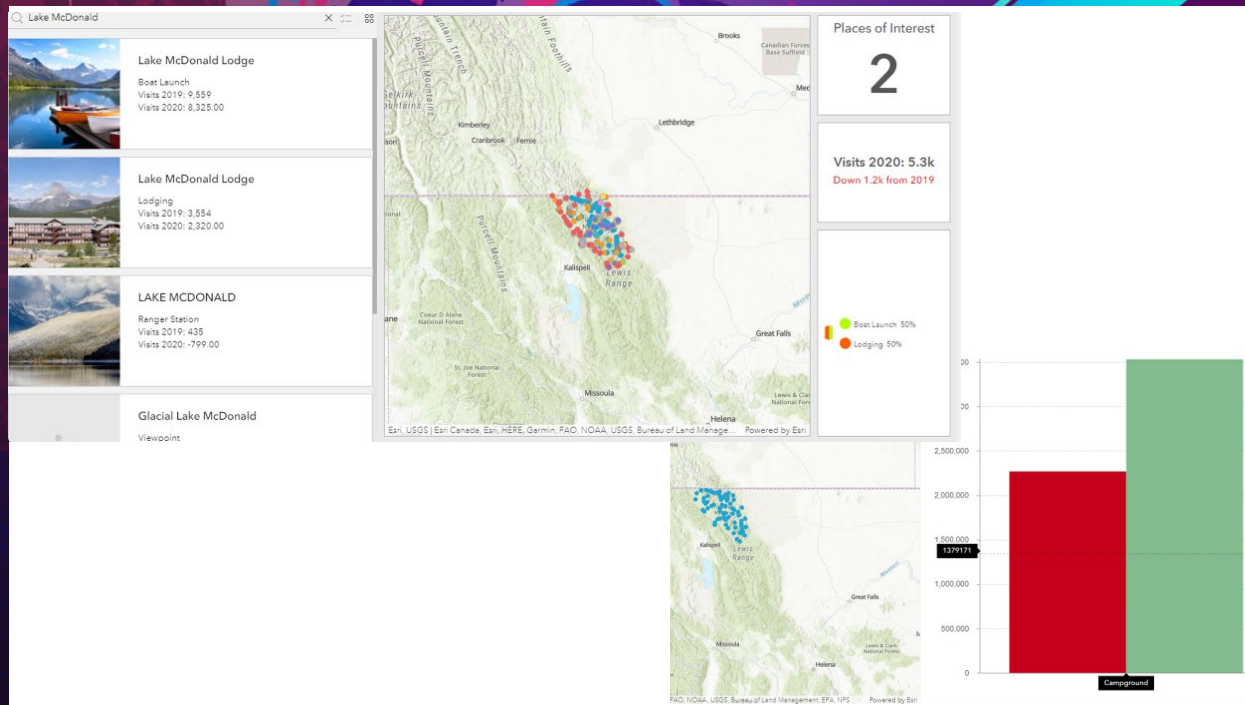
Kenyon Huber-Wilker

Integration With Other Apps

- With the Survey widget, you can pass attribute values and geometries of feature layers on the map to Survey123
- With the Embed widget, you can show Dashboards or dynamically
 - Display the dashboard with URL parameters from a field
 - Pass dynamic URLs from your web experience to the dashboard as URL parameters . For example
<https://experience.arcgis.com/experience/9b69e6e7a6984952944d0410ac9b1872/?state=California>
 - Apply to other apps as well. See the [blog](#) on the community

Dynamic URLs





Dynamic Dashboards & Widgets Output Data

Kenyon Huber-Wilker



Theme

Building Block

Theme

- Define a set of color and font
- Options to customize
 - Color
 - Font style
 - Font size

Default

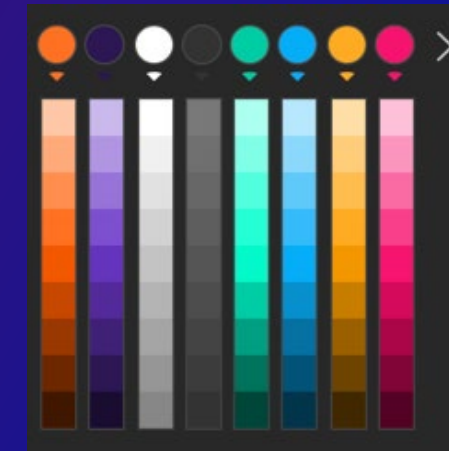
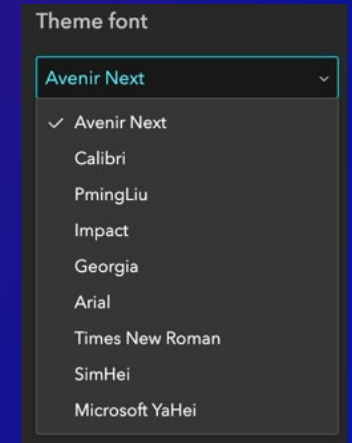
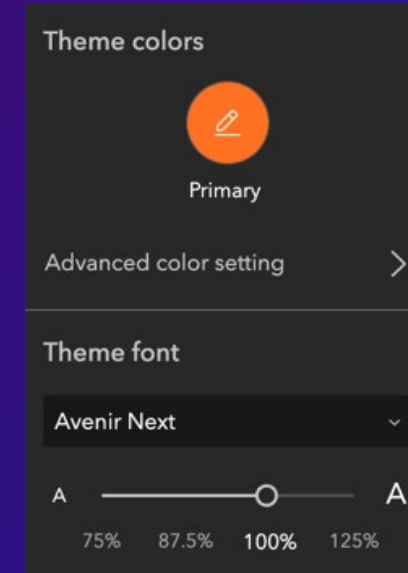
Dark

Vivid

New Ink

New Mysterious

Shared

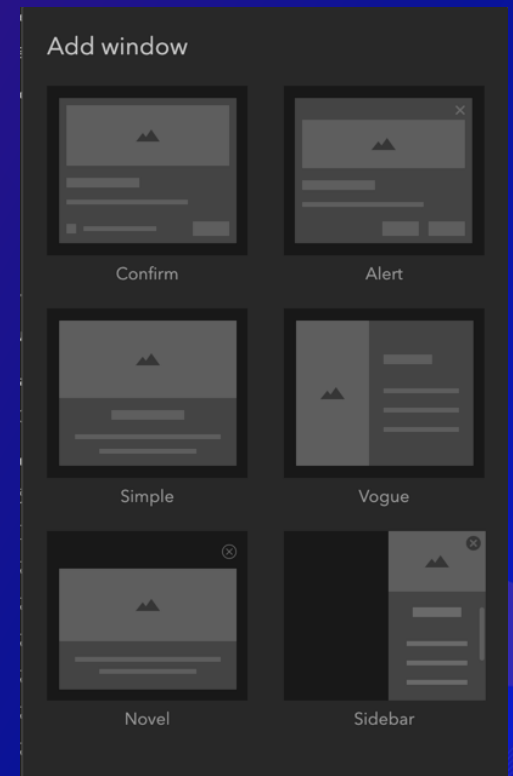
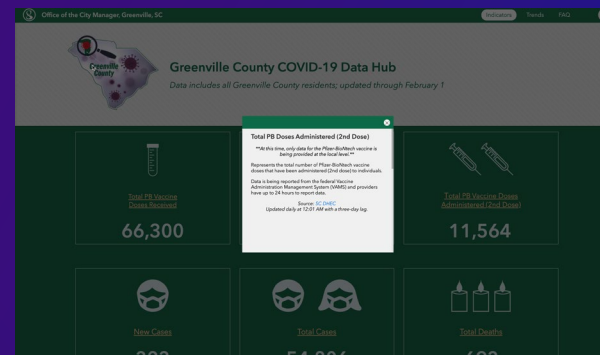
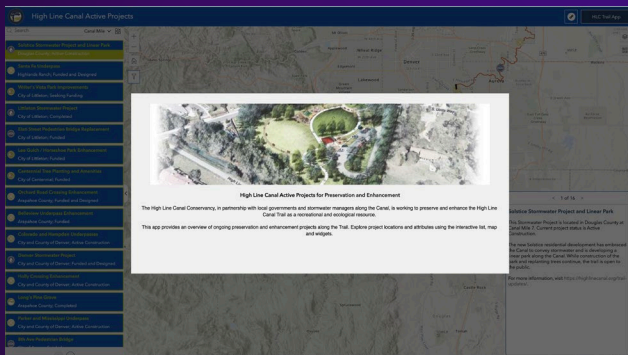


Window, Page, Screen Group

Building Block

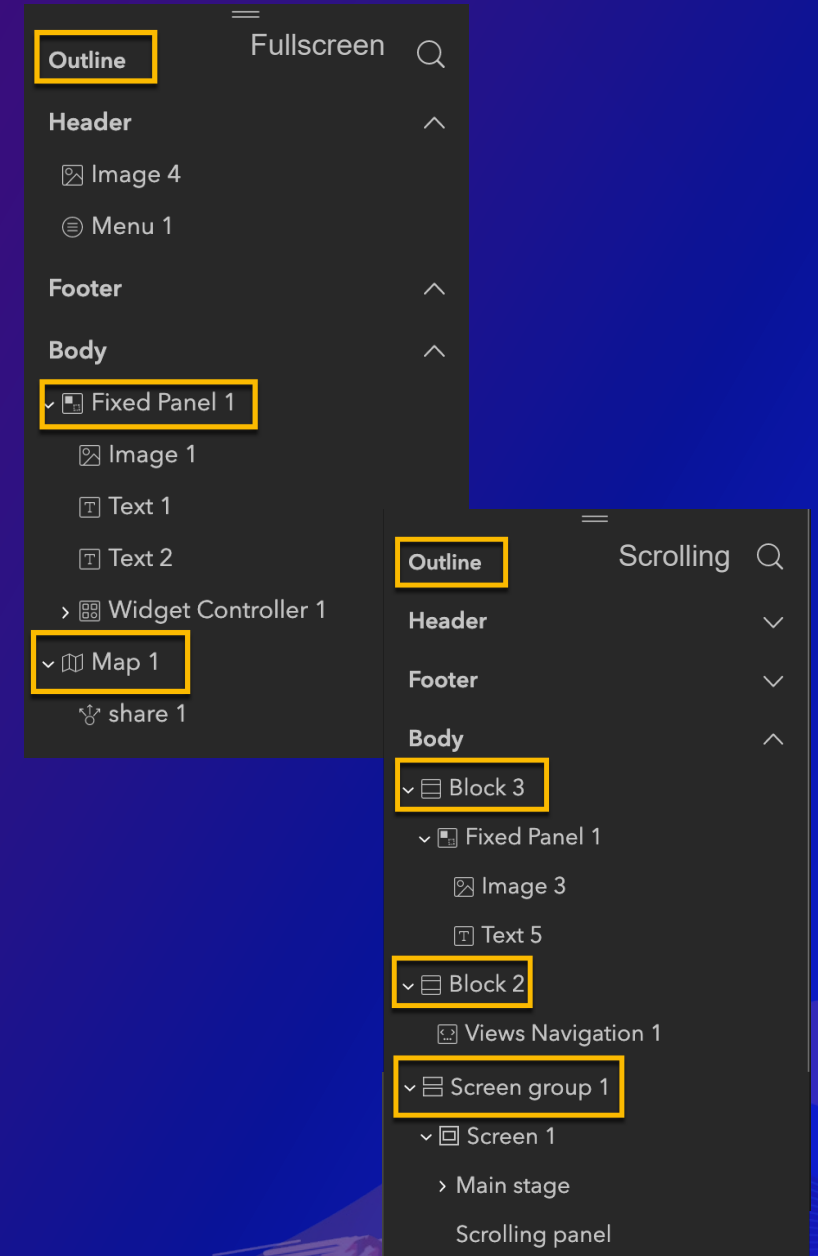
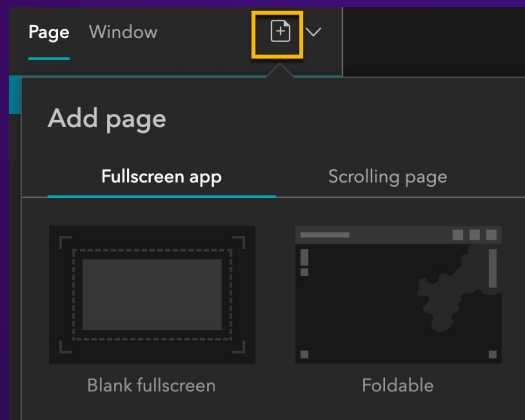
Window

- Splash screens, tips, information, confirmation, or alert window
- Two modes
 - Fixed: display in a fixed position like a splash screen
 - Anchored: display around the widget that triggers it
- Start with a window template
- Use any widgets that fit



Page

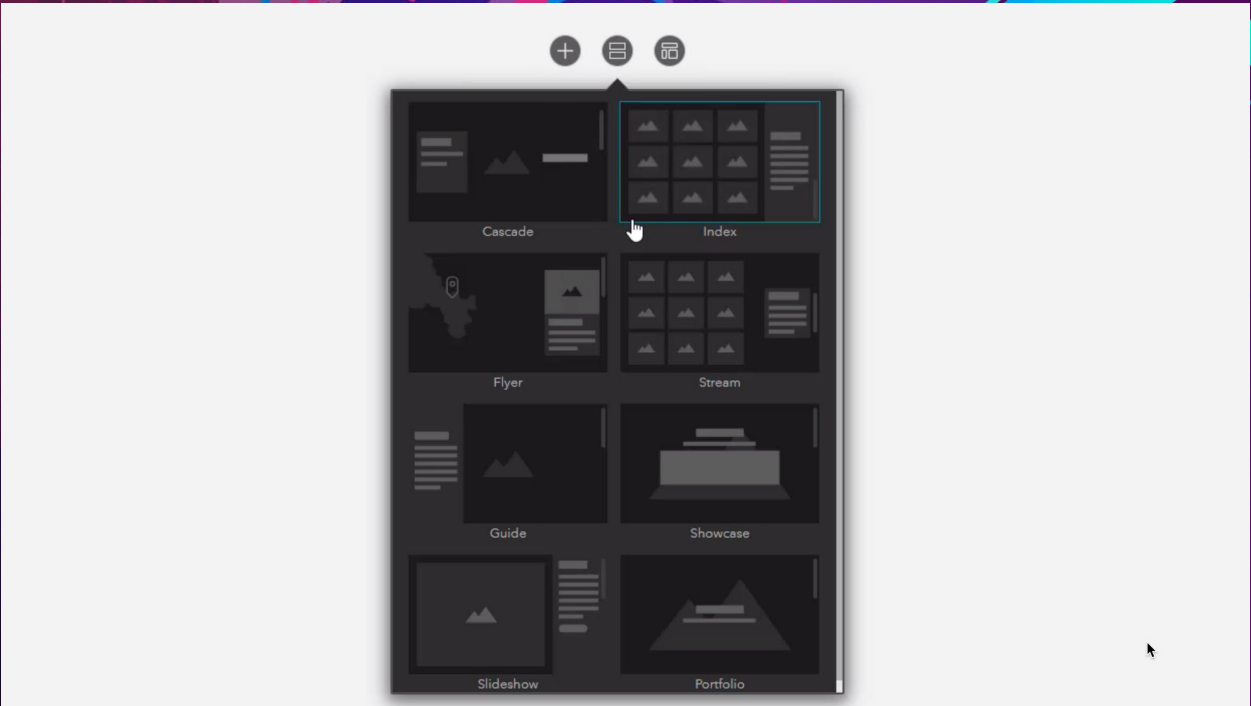
- Header, Footer, Open with window properties
- **Outline** helps navigate widgets on the page
 - The full-screen page is organized with widgets directly
 - The scrolling page is organized in Block and Screen group where widgets are nested
- Add new pages from default templates



Screen Group New

Engaging
Modern
Web experience





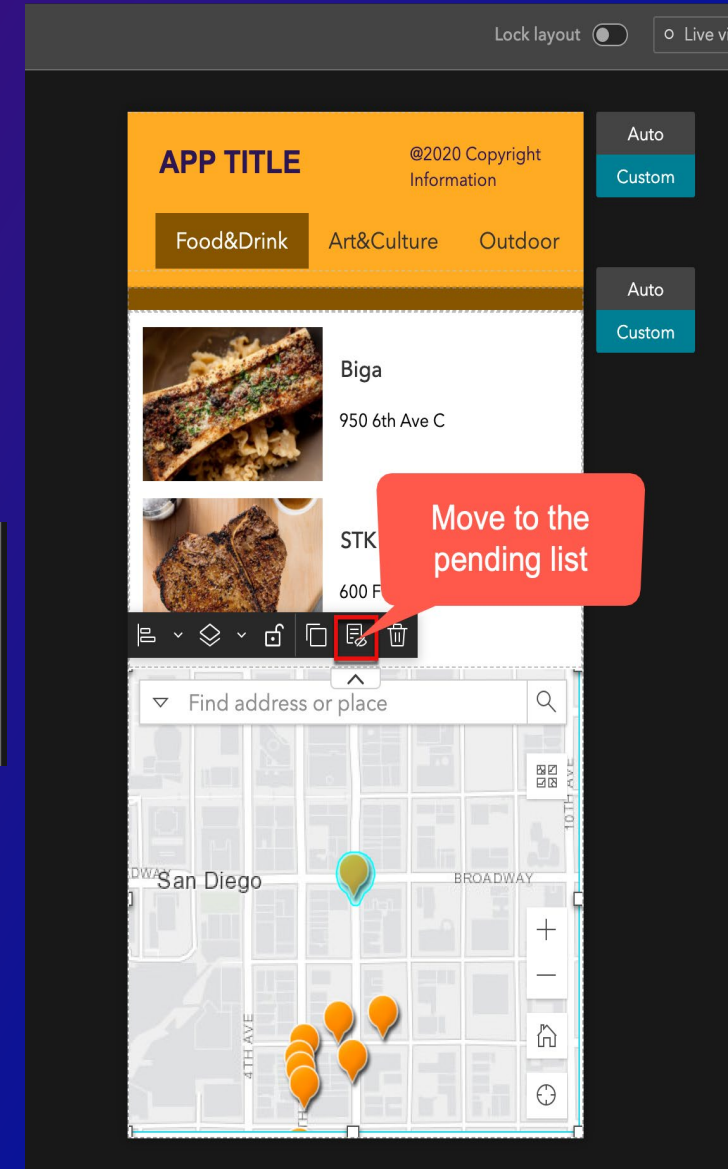
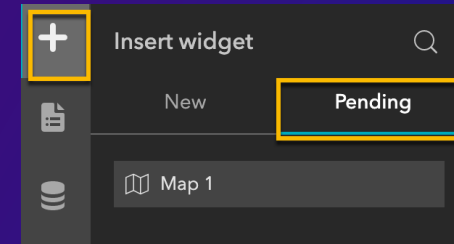
Screen Group Demo

Kenyon Huber-Wilker

Mobile Optimization

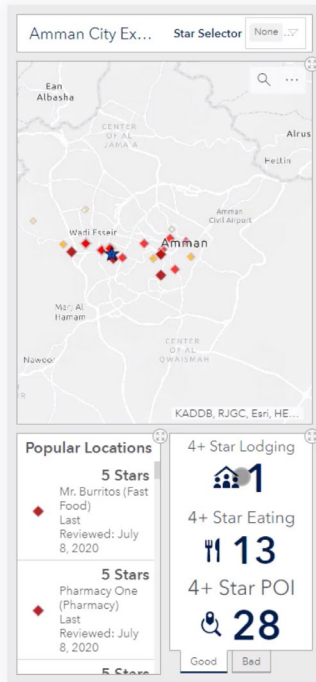


- You can configure an app on Mobile differently from that on Desktop with one URL, including
 - Change the content of a widget, such as
 - Turn off tools on the map
 - Change the font color and size
 - Embed the mobile version of a dashboard
 - Use less or different widgets
 - Change the layout
- The key is to move widgets to the pending list other than delete them, so they are available on the desktop



Mobile Optimization

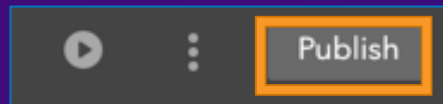
Kenyon Huber-Wilker



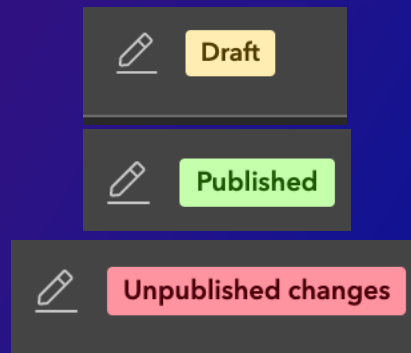
Publish Apps

The background is a vibrant, abstract composition. It features a deep purple field with large, flowing, organic shapes in shades of pink and orange. A thin, bright yellow arc curves across the right side of the image. In the bottom right corner, there is a stylized, high-contrast illustration of a mountain range or a rugged landscape in white and orange tones.

Publish Apps

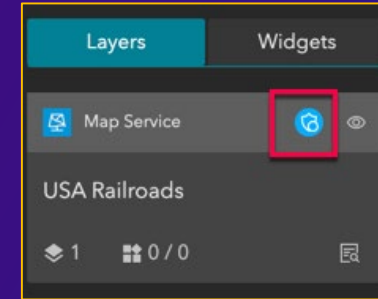


Draft
Published
Unpublished changes



Subscriber Content

Authorize access from
your account, so end
users are not asked to
sign in



Subscriber contents

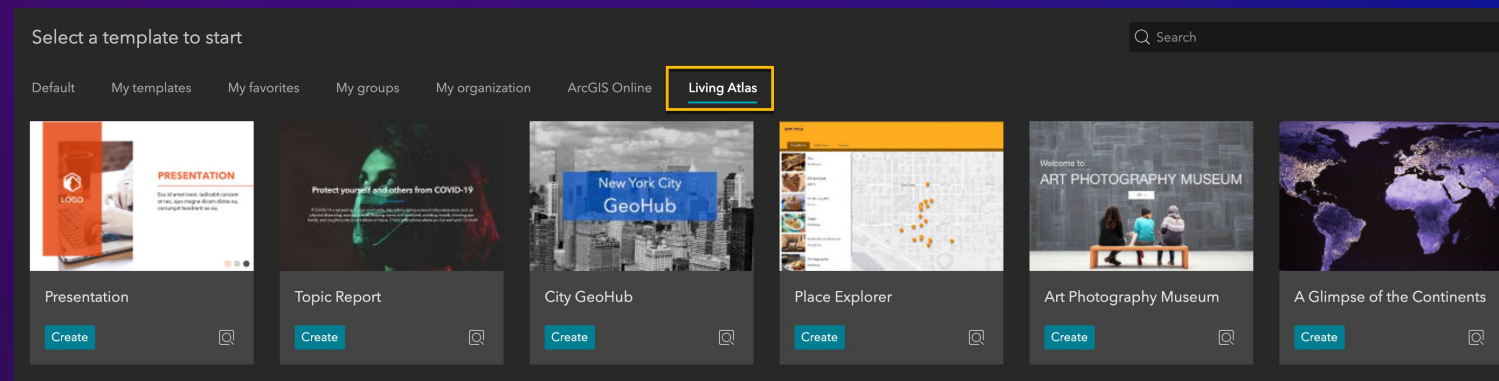
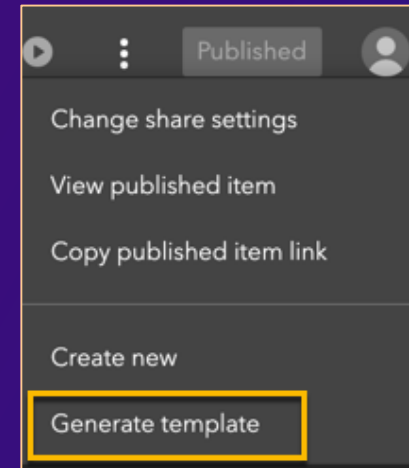
Certain resources used in this experience require an ArcGIS subscription. Enable public access by authoring with your subscription. Be aware that some content will consume credits.

Data

Authorized	Subscriber content	Credits	Request limit	Interval
☒	USA Railroads	✓		None

Generate templates

Create your own
starters for reuse
and branding
Shared templates



Roadmap

The background is a vibrant, abstract composition. It features a deep purple field with large, flowing, organic shapes in shades of pink and orange. A thin, bright yellow arc curves across the right side of the image. In the bottom right corner, there is a stylized, high-contrast illustration of a mountain range or a rugged landscape in white and orange tones.

Roadmap

Q3 2021

- Search
- Oriented Imagery

Near term

- Print
- Utility Network Trace
- Elevation Profile
- Business Analyst

Long term

- Accessibility
- Directions
- Add Data
- Time Slider
- Edit
- Sketch
- Geoprocessing
- Swipe
- Basemap Gallery
- Slice (3D)
- Daylight (3D)
- Line of Sight (3D)

Helpful Resources



ArcGIS Experience Builder Online Edition

experience.arcgis.com

Download Developer Edition

developers.arcgis.com/experience-builder

Help documentation

doc.arcgis.com

Product page

esri.com

Community

<https://community.esri.com/community/arcgis-experience-builder>

Sample Templates

Default My templates My favorites My groups My organization ArcGIS Online Living Atlas



Worldwide urban populati...

Create



Growth of urban catering i...

Create



Presentation

Create



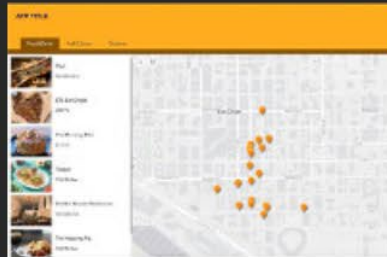
Topic Report

Create



City GeoHub

Create



Place Explorer

Create



Art Photography Museum

Create



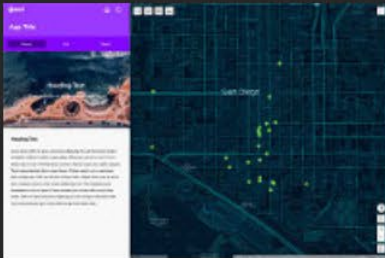
A Glimpse of the Continents

Create



National Flood Risk Asses...

Create



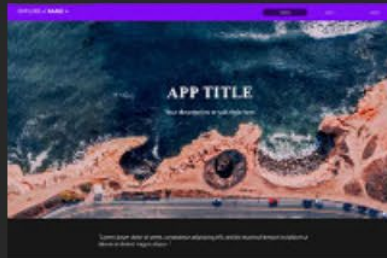
Map View

Create



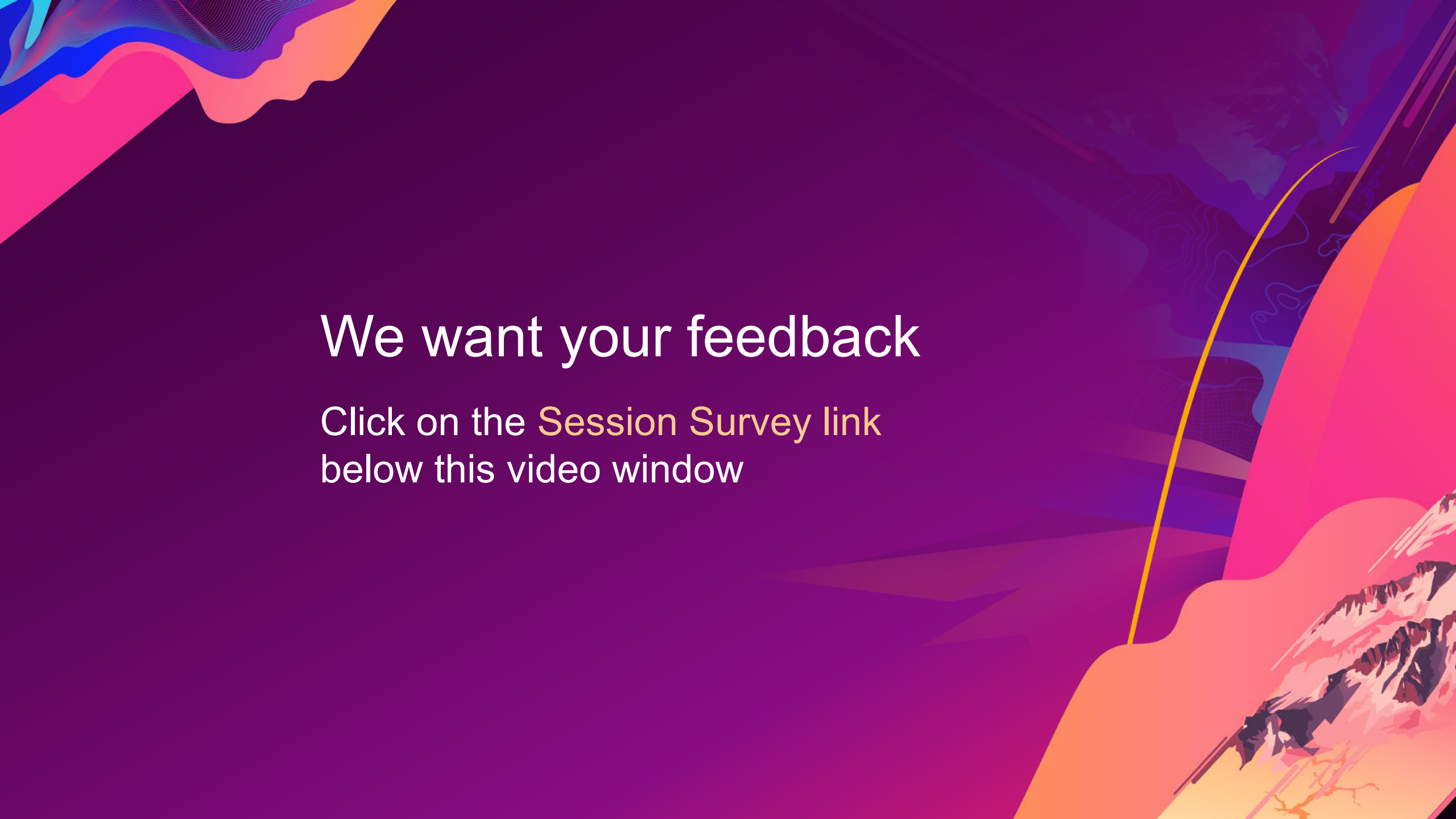
Public Info

Create



City Explorer

Create



We want your feedback

Click on the [Session Survey link](#)
below this video window



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